Computer Vision Course

School of Information and Communication Technology

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Computer Vision

Chapter 4: Self-supervising learning

Computer Vision Group
School of Information and Communications Technology

ONE LOVE. ONE FUTURE.

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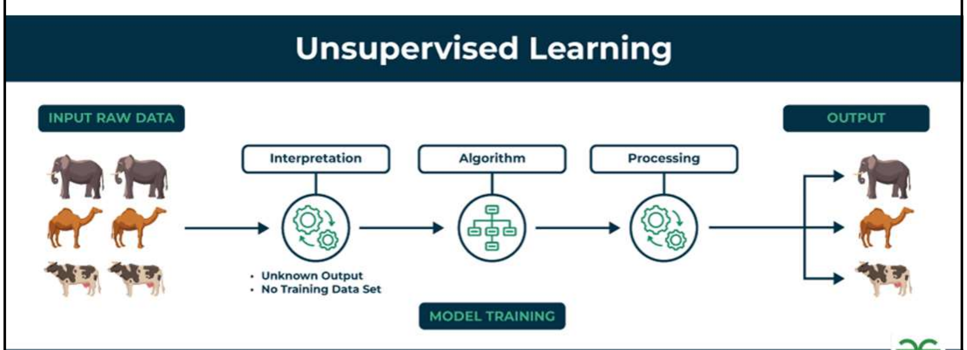
Contents

- Unsupervising and supervising learning
- Issues with unsupervising and supervising learning
- Self-supervising learning introduction
- Evaluation of self-supervising learning
- Pretext tasks

Un-supervising learning

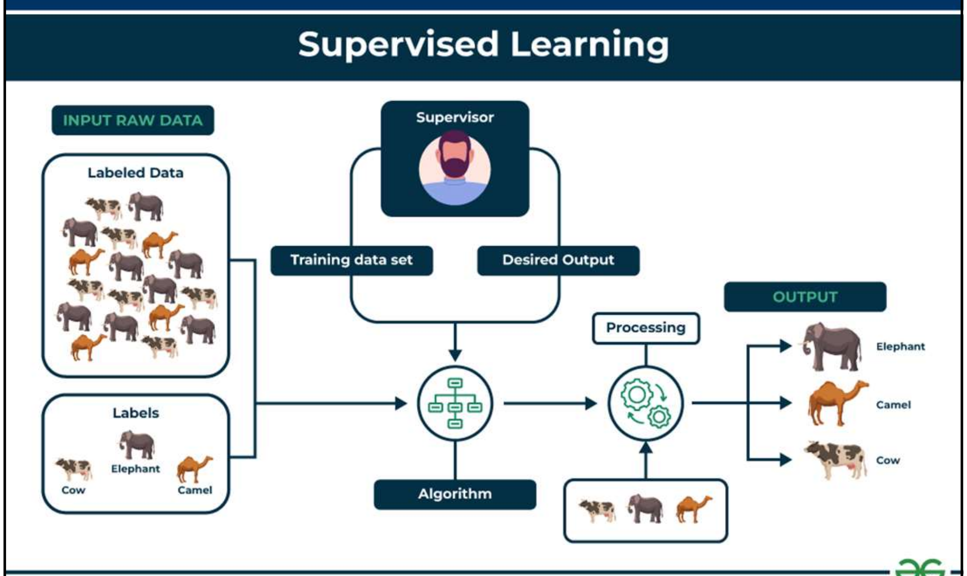
u im: không cn nhãn trc



Un-supervising learning

- Keypoints:
  - Unsupervised learning allows the model to discover patterns and relationships in unlabeled data.
  - Clustering algorithms group similar data points together based on their inherent characteristics.
  - Feature extraction captures essential information from the data, enabling the model to make meaningful distinctions.
  - Label association assigns categories to the clusters based on the extracted patterns and characteristics.
- Examples: clustering, dimensionality reduction, feature learning, density estimation, etc.

Supervised learning



Supervised learning

- Keypoints
  - Supervised learning involves training a machine from labeled data.
  - Labeled data consists of examples with the correct answer or classification.
  - The machine learns the relationship between inputs (fruit images) and outputs (fruit labels).
  - The trained machine can then make predictions on new, unlabeled data.
- Examples: Classification, regression

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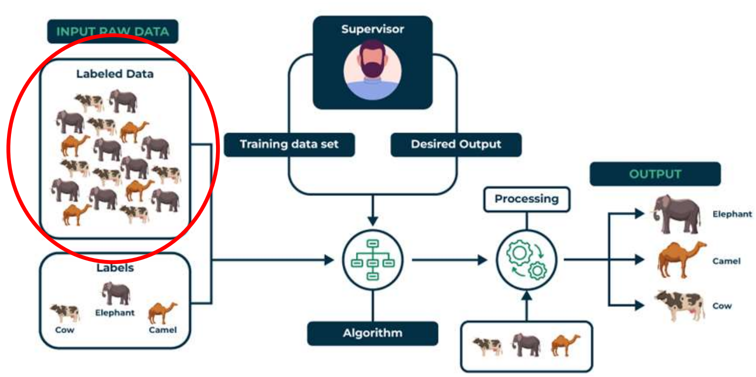
Problem of supervised learning?

- The need for labeled data
- Time and cost consuming



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Example of expensive annotation

- Assume you want to label 1M images. How much will it cost?
  - 1000000 images – small to medium sized dataset
  - 10 seconds/image (fast)
  - $15/hour wage (low paid)
- What if we need 1B images to be labelled?

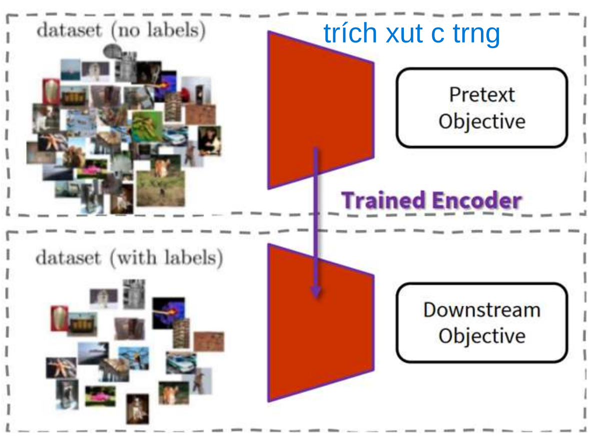
Supervised learning is not HOW WE LEARN

Babies don't get supervision for everything they see!

Some solutions for the problem

- Lets build methods that learn from ”raw” data – no annotations required
- **Unsupervised Learning**: Model isn’t told what to predict. Older terminology, not used as much today.
- **Self-Supervised Learning**: Model is trained to predict some naturallyoccurring signal in the raw data rather than human annotations
- **Semi-Supervised Learning**: Train jointly with some labeled data and (a lot) of unlabeled data.

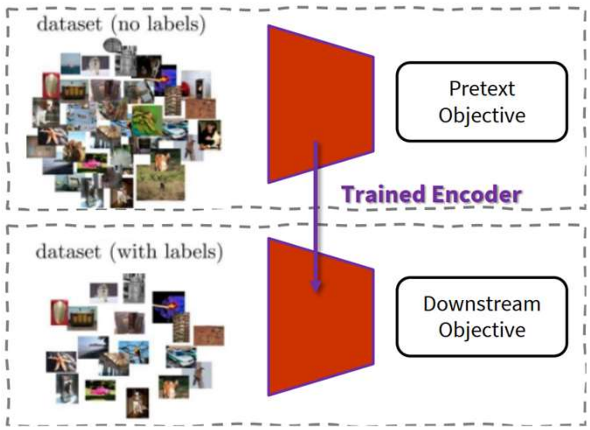
Self-Supervised Learning



Pretext Task

- Define a task based on the data itself
- No manual annotation
- Could be considered an unsupervised task;
- but we learn with supervised learning objectives, e.g., classification or regression.

Self-Supervised Learning



Downstream task

- The application you care about
- You do not have large datasets
- The dataset is labeled

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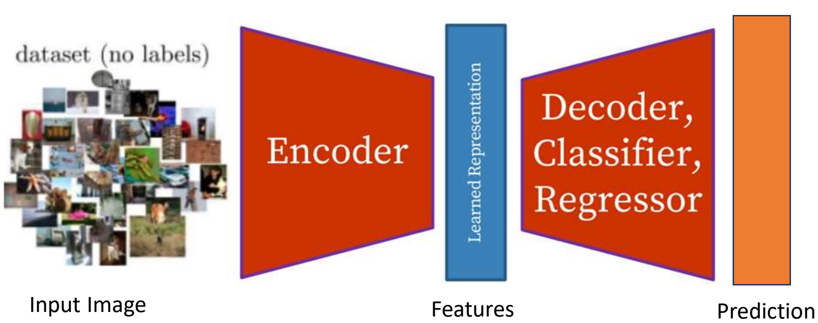
Self-Supervised Learning: Pretext then Transfer

- Labels/outputs automatically generated from data



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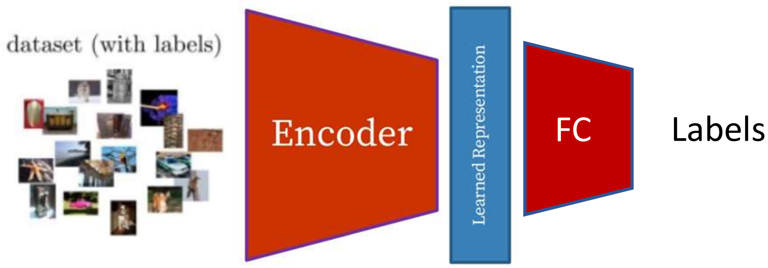
Self-Supervised Learning – Downstream Task



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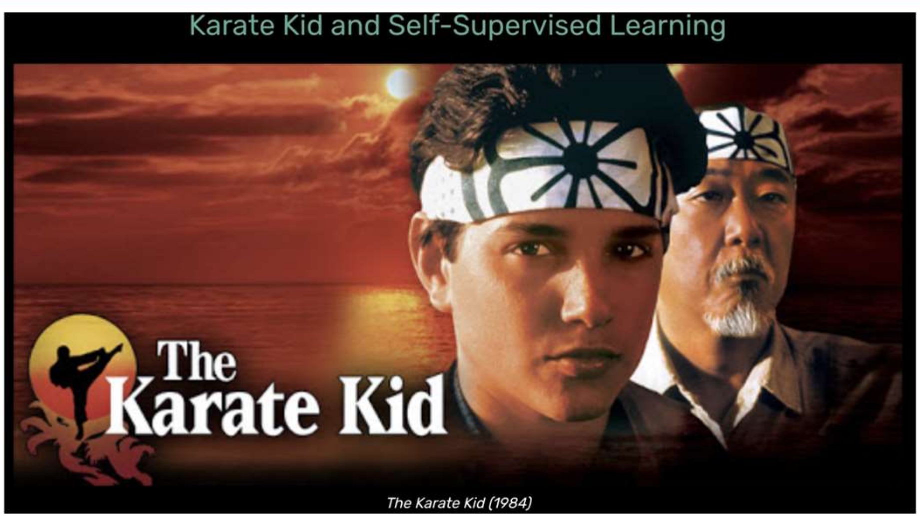
Example of real-life self-supervising learning



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Example of real-life self-supervising learning

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Example of real-life self-supervising learning

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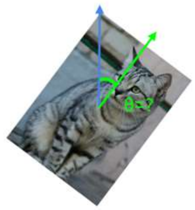
Self-supervised pretext tasks

- Example: learn to predict image transformations/ complete corrupted images

image completion



rotation prediction

"jigsaw puzzle"

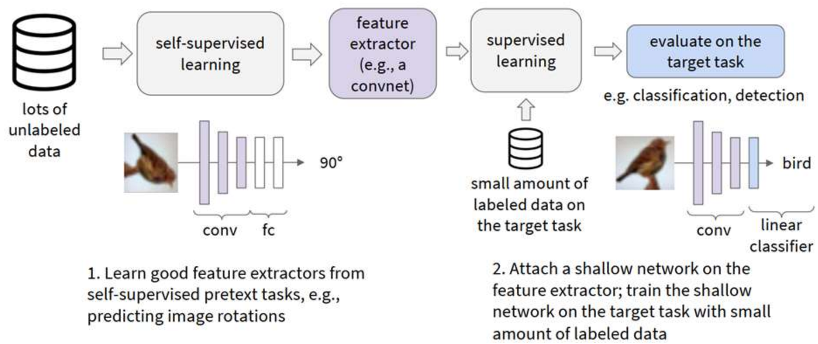
colorization

1. Solving the pretext tasks allow the model to learn good features
2. We can automatically generate labels for the pretext tasks

How to evaluate a self-supervised learning method?

- Pretext Task Performance
  - Measure how well the model performs on the task it was trained on without labels.
- Representation Quality: Evaluate the quality of the learned representations
  - Linear Evaluation Protocol: Train a linear classifier on the learned representations;
  - Clustering: Measure clustering performance;
  - t-SNE: Visualize the representations to assess their separability.)
- Robustness and Generalization
  - Test how well the model generalizes to different datasets and is robust to variations.
- Computational Efficiency
  - Assess the efficiency of the method in terms of training time and resource requirements.
- Transfer Learning and Downstream Task Performance
  - Assess the utility of the learned representations by transferring them to a downstream supervised task

How to evaluate a self-supervised learning method



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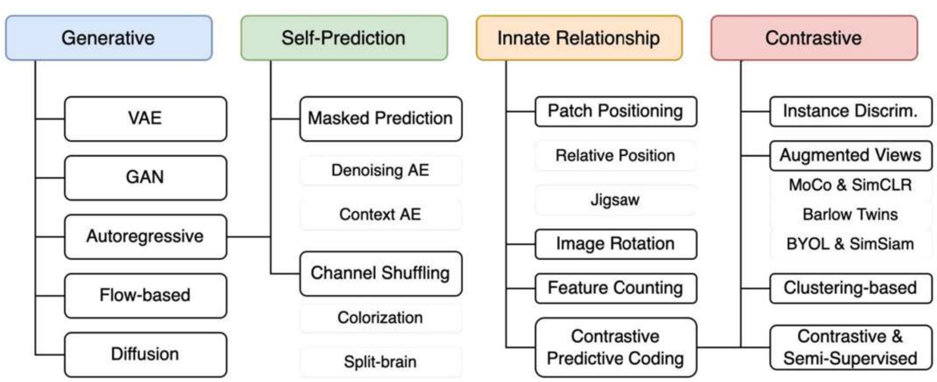
Pretext task(s)



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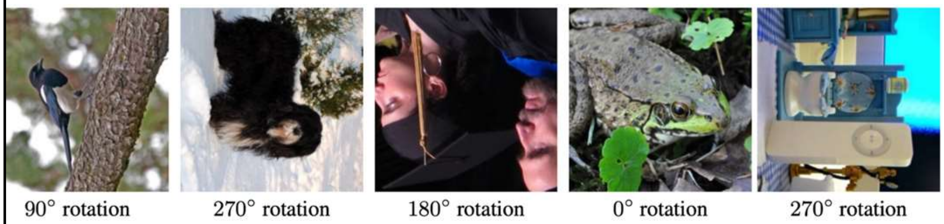
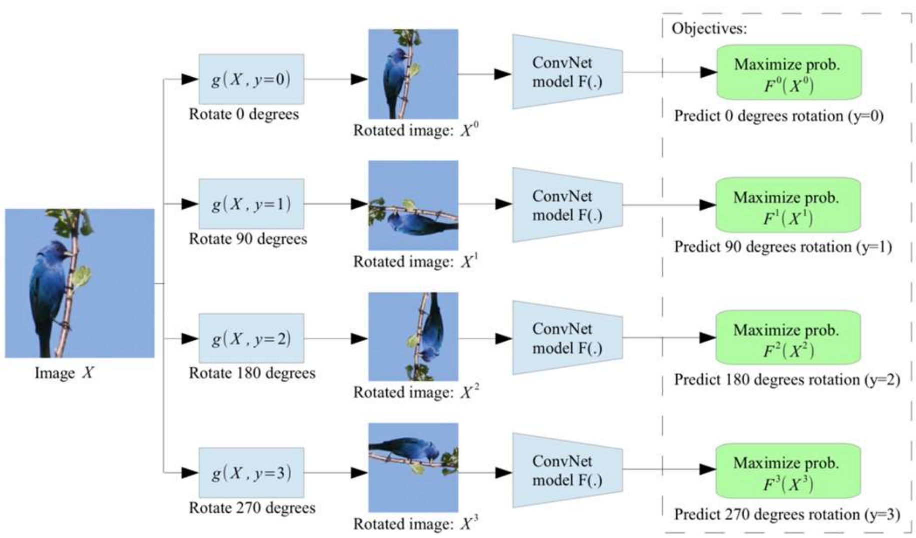
Pretext task: Rotation prediction



- Hypothesis: a model could recognize the correct rotation of an object only if it has the "visual commonsense" of what the object should look like unperturbed
  - Much easier if you recognize the content

Pretext task: Rotation prediction



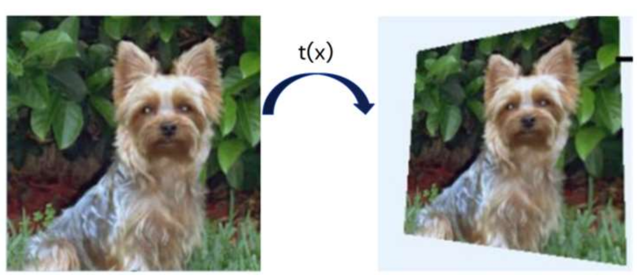
Pretext task: Rotation prediction

- Pros
  - Very simple to implement and use, while being quite effective
- Cons
  - Assumes training images are photographed with canonical orientations (and canonical orientations exist)
  - Train-eval gap: no rotated images at eval
  - Not fine-grained enough due to no negatives from other images
    - e.g. no reason to distinguish cat from dog
  - Small output space - 4 cases (rotations) to distinguish [not trivial to increase; see later]
  - Some domains are trivial e.g. StreetView ⇒ just recognize sky

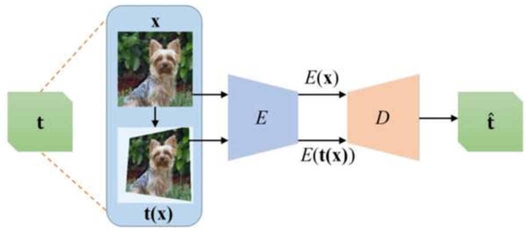
Pretext task: relateive transformation prediction

- Estimate the transformation between two images.

→ Requires good features



Pretext task: relateive transformation prediction

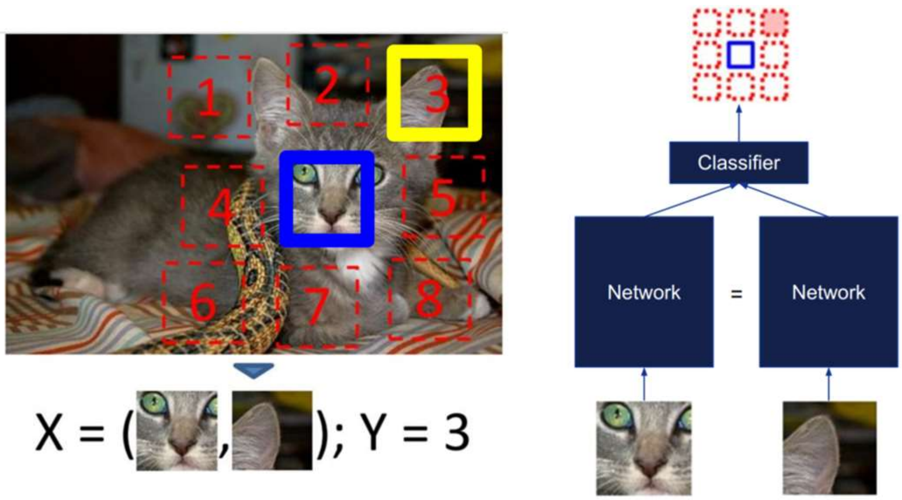


- Pros
  - In line with classical computer vision, e.g. SIFT was developed for matching
- Cons
  - Train-eval gap: no transformed images at eval
  - Not fine-grained enough due to no negatives from other images
    - e.g. no reason to distinguish cat from dog
  - Questionable importance of semantics vs low-level features (assuming we want semantics)
    - Features are potentially not invariant to transformations

Pretext task: context prediction

- Can you guess the spatial configuration for the two pairs of patches
  - Much easier if you recognize the object!

Pretext task: context prediction

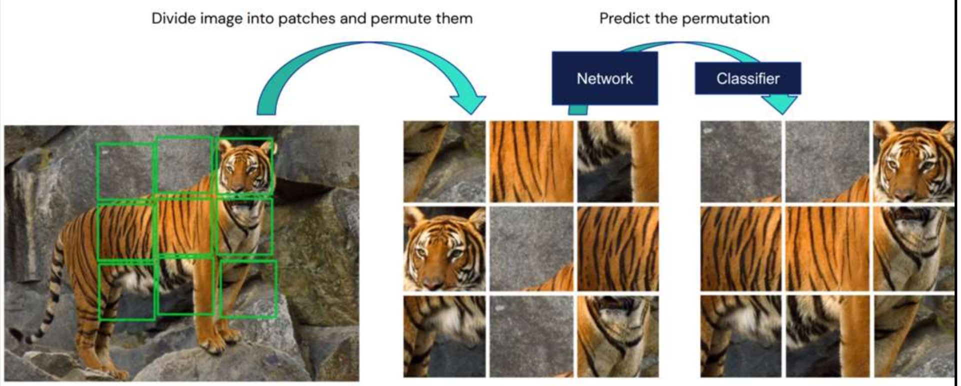


Pretext task: context prediction

- Pros
  - (arguably) The first self-supervised method
  - Intuitive task that should enable learning about object parts
- Cons
  - Assumes training images are photographed with canonical orientations (and canonical orientations exist)
  - Training on patches, but trying to learn image representations
  - Not fine-grained enough due to no negatives from other images
    - e.g. no reason to distinguish cat from dog eyes
  - Small output space - 8 cases (positions) to distinguish?

Pretext task: jigsaw puzzles

- Pros & Cons: Same as for context prediction apart from being harder



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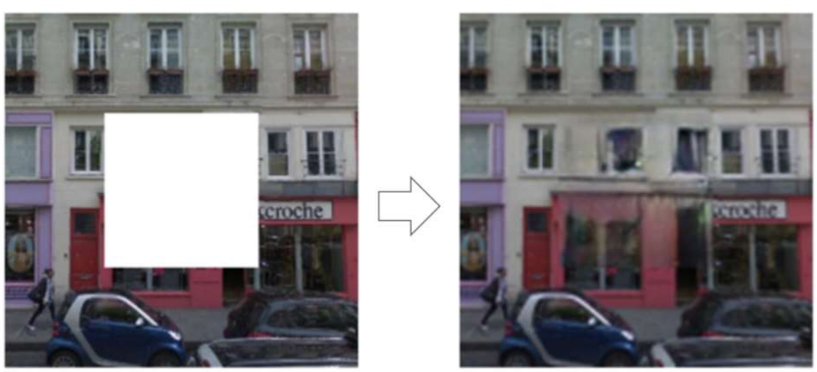
Pretext tasks: predict missing pixels

- Context Encoders: Feature Learning by Inpainting
  - What goes in the middle? Much easier if you recognize the objects!



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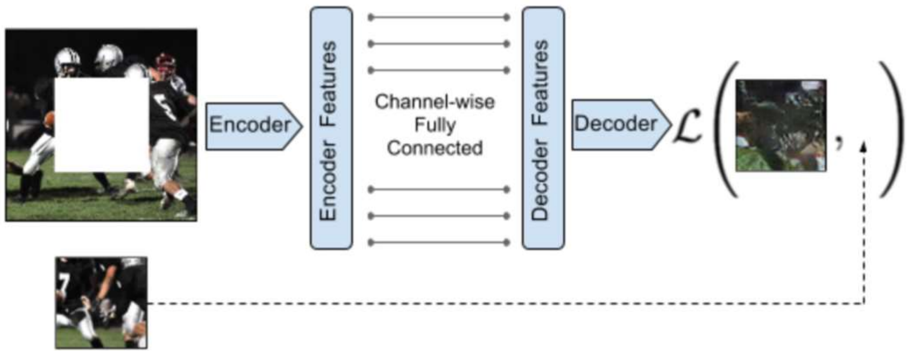
Pretext tasks: predict missing pixels

- Learning to reconstruct the missing pixels



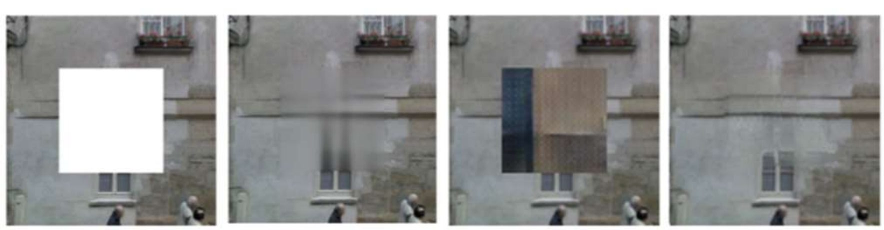
Pretext tasks: predict missing pixels

Loss = reconstruction + adversarial learning

$$L(x) = L_{recon}(x) + L_{adv}(x)$$

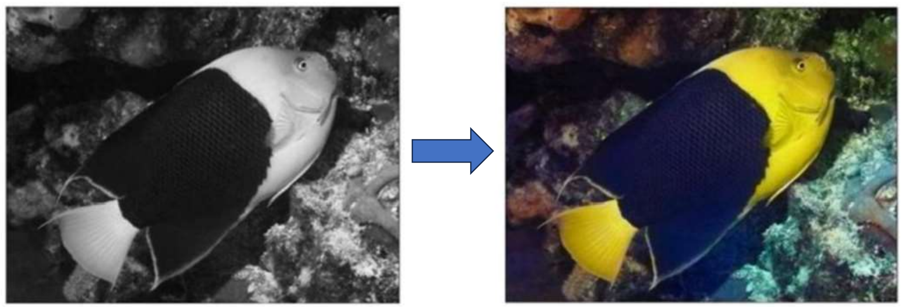
$$L_{recon}(x) = ||M * (x - F_\theta((1 - M) * x))||_2^2$$

$$L_{adv} = \max_D \mathbb{E}[\log(D(x))] + \log(1 - D(F((1 - M) * x)))]$$



Pretext tasks: colorization

- What is the colour of every pixel?
  - Hard if you don't recognize the object!



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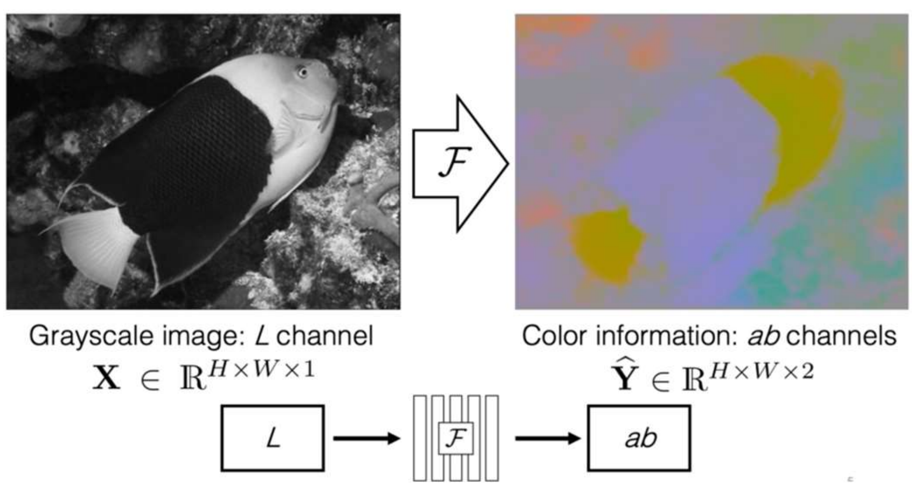
Pretext tasks: colorization



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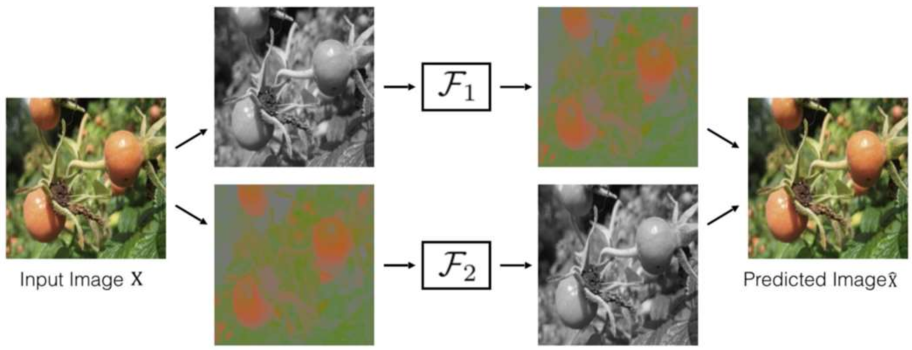
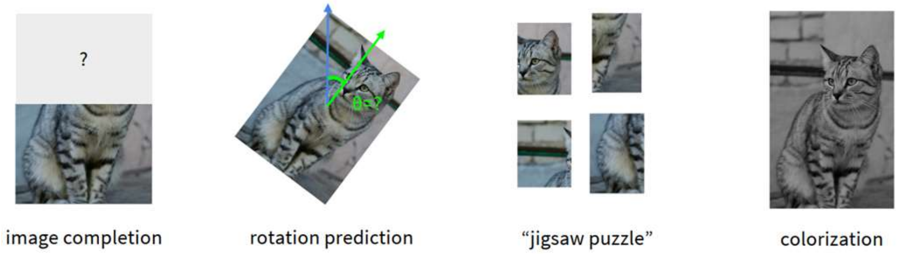
Pretext tasks: Splitted brain



Summary: pretext tasks from image transformations

- Pretext tasks focus on "visual common sense", e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We don't care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- Problems: 1) coming up with individual pretext tasks is tedious, and 2) the learned representations may not be general

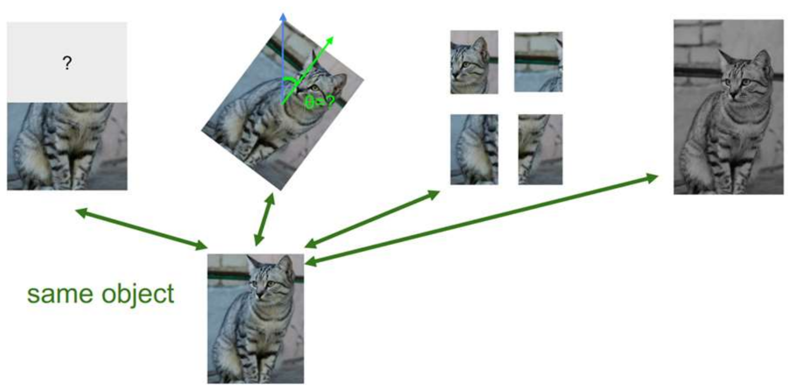
Pretext tasks from image transformations



Learned representations may be tied to a specific pretext task!

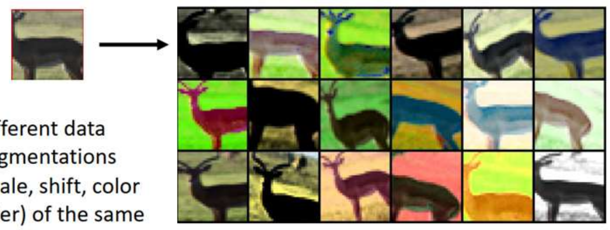
Can we come up with a more general pretext task?

A more general pretext task?



Exemplar CNN

Quiz: What is this?



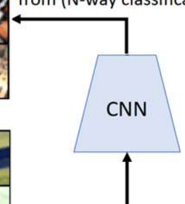
Different data augmentations (scale, shift, color jitter) of the same initial image patch

Answer: Deer!

Exemplar CNN

Given an initial dataset of N image patches

Predicts which of the N original images it came from (N-way classification)

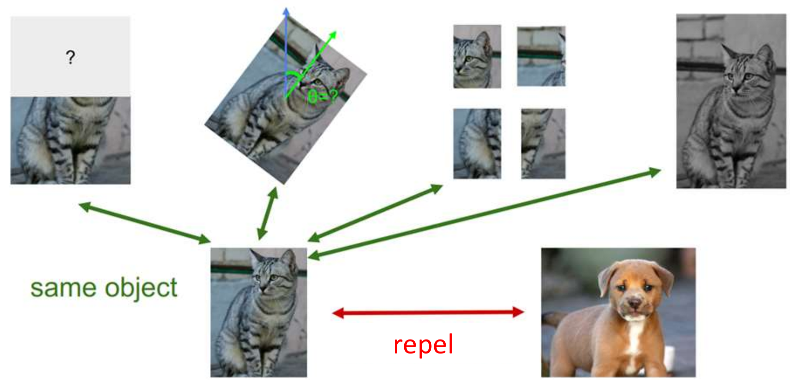


CNN inputs an augmented patch

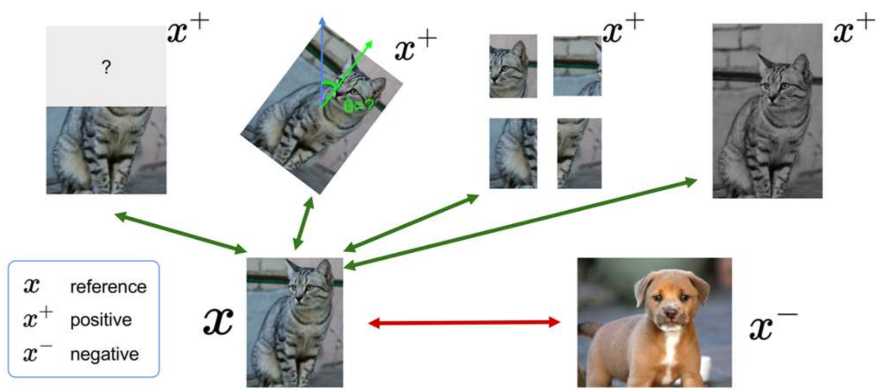
Sample K different augmentations for each; now have K*N total patches

Problem: number of parameters in final layer depends on N; hard to scale

Contrastive Representation Learning



Contrastive Representation Learning



A formulation of contrastive learning

- What we want:

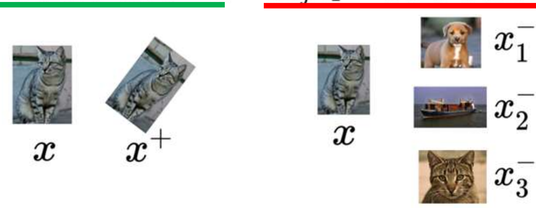
$$\text{score}(f(x), f(x^+)) >> \text{score}(f(x), f(x^-))$$

- x: reference sample; x+ positive sample; x- negative sample
- Given a chosen score function, we aim to learn an encoder function f that yields high score for positive pairs (x, x+) and low scores for negative pairs (x, x-).

A formulation of contrastive learning

- Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+))}{\exp(s(f(x), f(x^+)) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-))} \right]$$



A formulation of contrastive learning

- Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+)))}{\underbrace{\exp(s(f(x), f(x^+)))}_{\text{score for the positive pair}} + \underbrace{\sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))}_{\text{score for the N-1 negative pairs}}} \right]$$

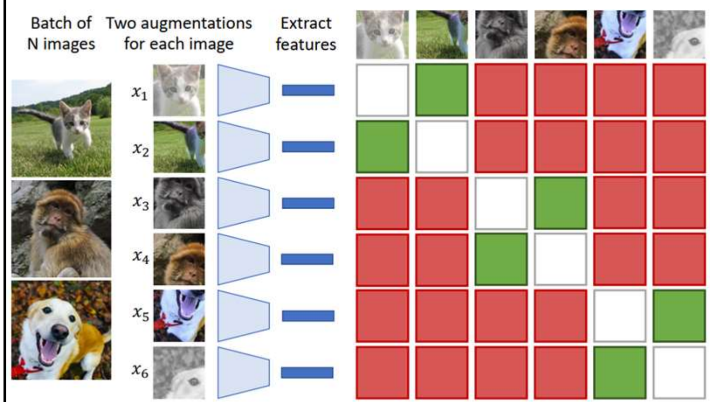
- This seems familiar …
- Cross entropy loss for a N-way softmax classifier!
  I.e., learn to find the positive sample from the N samples
- Commonly known as the InfoNCE loss
- A *lower bound* on the mutual information between *f(x)* and *f(x+)*

$$MI[f(x), f(x^+)] - \log(N) \geq -L$$

- The larger the negative sample size (*N*), the tighter the bound

SimCLR: A Simple Framework for Contrastive Learning

SimCLR: A Simple Framework for Contrastive Learning



Each image tries to predict which of the *other* 2N-1 images came from the same original image

Similarity between x_i and x_j:

$$s_{i,j} = \frac{\phi(x_i)^T\phi(x_j)}{||\phi(x_i)|| \cdot ||\phi(x_i)||}$$

If (x_i, x_j) is a positive pair, then loss for x_i is:

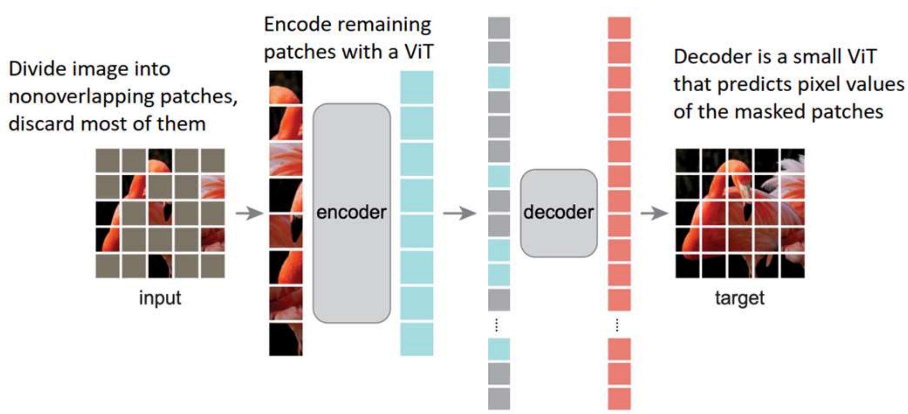
$$L_i = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1, k\neq i}^{2N} \exp(s_{i,k}/\tau)}$$

(τ is a *temperature*)

Interpretation: Cross-entropy loss over the other 2N-1 elements in the batch!

Masked Autoencoders (MAE)

- A method dethrones contrastive learning? Denoising Autoencoder with Vision Transformer



Masked Autoencoders (MAE): Reconstructions

Input Patches Prediction Actual Image

Multimodal Self-Supervised Learning

- Don't learn from isolated images -- take images together with some context

Video: Image together with adjacent video frames

Agrawal et al, "Learning to See by Moving", ICCV 2015
Wang et al, "Unsupervised Learning of Visual Representations using Videos", ICCV 2015
Pathak et al, "Learning Features by Watching Objects Move", CVPR 2017

Sound: Image with audio track from video

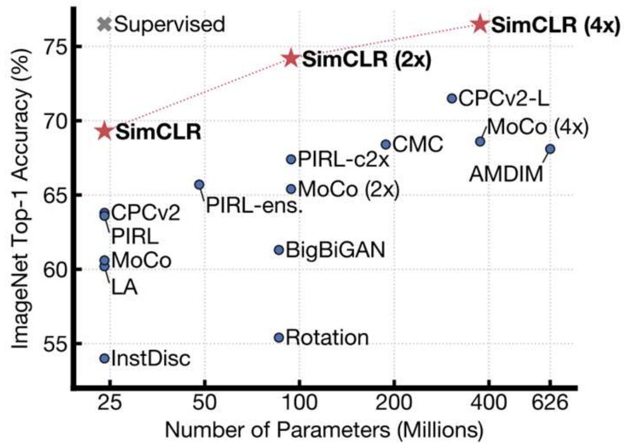
Owens et al, "Ambient Sound Provides Supervision for Visual Learning", ECCV 2016
Arandjelovic and Zisserman, "Look, Listen and Learn", ICCV 2017

3D: Image with depth map or point cloud

Xie et al, "PointContrast: Unsupervised Pre-training for 3D Point Cloud Understanding", ECCV 2020
Zhang et al, "Self-supervised pretraining of 3D features on any point-cloud", CVPR 2021

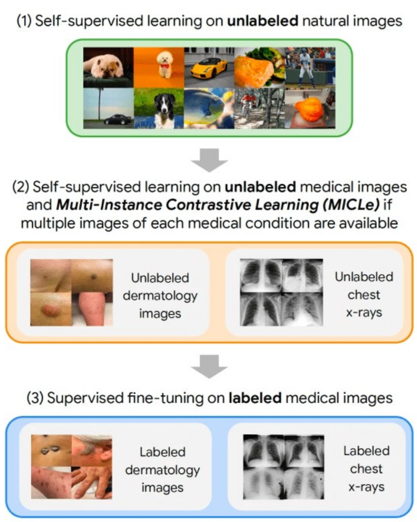
Language??

Some results of using self-supervised learning

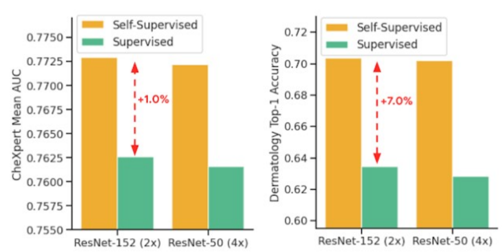


ImageNet Top-1 accuracy of linear classifiers trained on representations learned with different self-supervised methods (pretrained on ImageNet).

Some results of using self-supervised learning

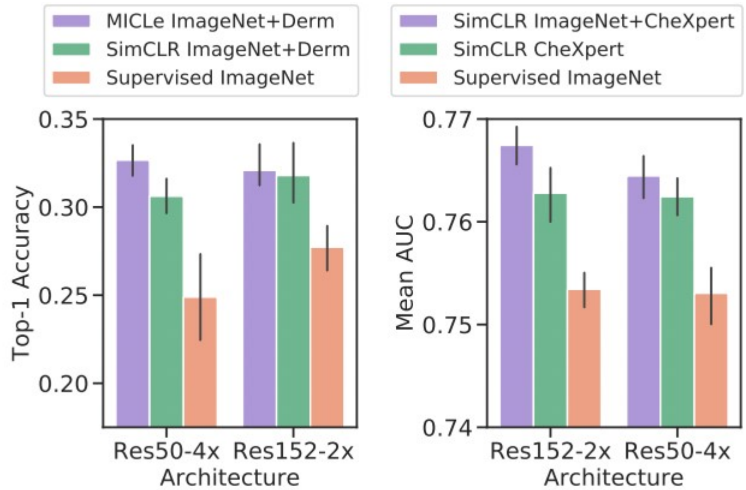


- Google's medical imaging analysis model
  - Step1: SimpleCLR



Comparison of supervised and self-supervised pre-training, followed by supervised fine-tuning using two architectures on dermatology and chest X-ray classification

Some results of using self-supervised learning



Performance of solutions for
Google's medical imaging analysis model

Challenges in self-supervised learning

- Accuracy
- Computational Efficiancy
- Pretext task

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THANK YOU !

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